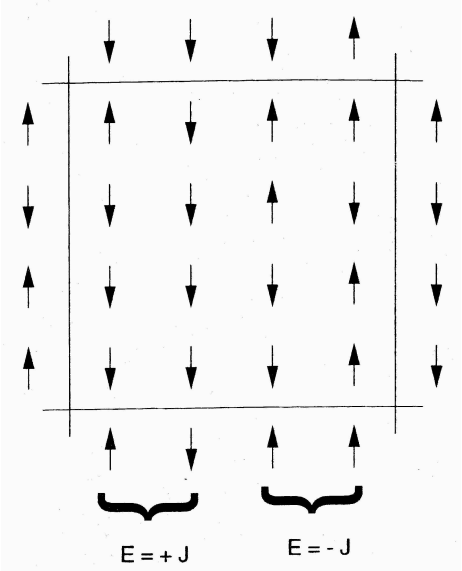
Ising model



- Random processes of many particles
- The ***interaction*** between particles play an essential role
- Many interacting particles can display ***phase transitions***
- Examples: melting of a solid, ferromagnetism,…
- The concept of **temperature** is important for these examples → thermal and statistical physics
- **Magnetism** is a quantum phenomenon; each electron has a ***spin*** and the inherent magnetic moment
- Ferromagnetism arises when all spins point to the same direction → total magnetic moment is macroscopic in size
- Central issue is how the spins interact with each other and how this leads to the overall alignment; **temperature-dependence** is important
- Our basic model for a **ferromagnet** considers spins (arrows) placed in a regular lattice; each spin points either up or down, $s_i = \pm 1$

- Each spin interacts with the other spins, **Ising spins**
- Parallel alignment favorable for a ferromagnet (and reverse for an antiferromagnet)
- For a simple **nearest-neighbor interaction** the energy has a simple form

$$E = -J \sum_{\langle ij \rangle} s_i s_j$$

- The sum <*ij*> goes over all neighboring spin pairs, *J* is known as the **exchange constant** (positive)
- Ising spins with the given energy define the ***Ising model***
- The energy is lowest for spontaneous magnetization (parallel spins), and it is higher for disordered systems (temperature effect)
- The system is coupled to a **heat path**, ***canonical ensemble***; spins flip between different configurations (*T* effect)
- The probability of finding the system in a given configuration is proportional to the ***Boltzmann factor***

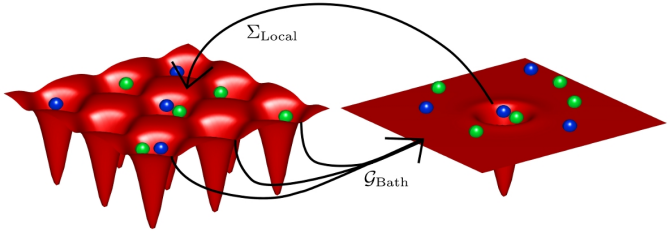
$$P_\alpha \sim e^{-E_\alpha / k_B T}$$

- Energy E_α corresponds to a particular configuration of spins, ***microstate***
- For a lattice of N spins there are 2^N different possible microstates of the system (N large → number of microstates is **very large**)
- The system interacts with a heath path and this causes individual spins to flip (energy gain/reduction); the total magnetic moment (macroscopic measurement) effectively averages over the microstates the system visits
- We need to calculate **probabilities for the system in various microstates**
- The ***measured*** magnetization will be then

$$M = \sum_\alpha M_\alpha P_\alpha$$

- Similarly, other macroscopic properties can be expressed via P_α
- The model above is ***extremely simple***, and many others exits
- For example, the **XY model (2D)** and the **Heisenberg model (3D)** with freely rotating spins, longer interaction lengths (second neighbors, etc.), quantum mechanical treatment…
- Yet, the simple Ising model captures many essential features of the phase transition to the ferromagnetic state

Mean field theory



- We consider next an **approximate** approach, ***mean field theory***, which will provide us some useful but not quantitative results for a spin system
- The magnetization is closely related to the **average spin alignment <s_i>**; thermal average over all microstates that are generated by spin flips
- For an infinite system, all spins have the same average alignment; spins equivalent, each have four nearest neighbors (square lattice)
- The **total magnetization** is

$$M = \sum_i \langle s_i \rangle = N \langle s_i \rangle$$

- An exact calculation of <s_i> requires the probabilities of all microstates; instead we use an approximation based on **the mean field theory, MFT**
- Let us add a **magnetic field** *H* to the problem, which tends to align the spins parallel

$$E = -J \sum_{\langle ij \rangle} s_i s_j - \mu H \sum_i s_i$$

- Let us assume first that we have **a single spin system**, which has two possible states $s_i = \pm 1$ and energies $E_\pm = -/+\mu H$ (← note the order)
- The **probabilities** to find the system in either of these states is (C is a constant which we can determine)

$$P_+ = Ce^{+\mu H/k_B T}$$
$$P_- = Ce^{-\mu H/k_B T}$$

- Thermal average is then (exact result, **check**)

$$\langle s_i \rangle = \sum_{s_i=\pm 1} s_i P_\pm = P_+ - P_- = \tanh(\mu H/k_B T)$$

- We use the previous result for obtaining an **approximate solution** for N interacting spins; MFT is based on an assumption that each spin sees the same ***effective magnetic field H_{eff}***
- The energy function can be written for a single spin in a suggestive form

$$E = -J\left(\sum_{\langle ij \rangle} s_j\right) s_i - \mu H s_i$$

- **Approximation**: Assume that spin variables s_j can be replaced by their thermal averages, this is the same for all spins and denoted by $\langle s \rangle$

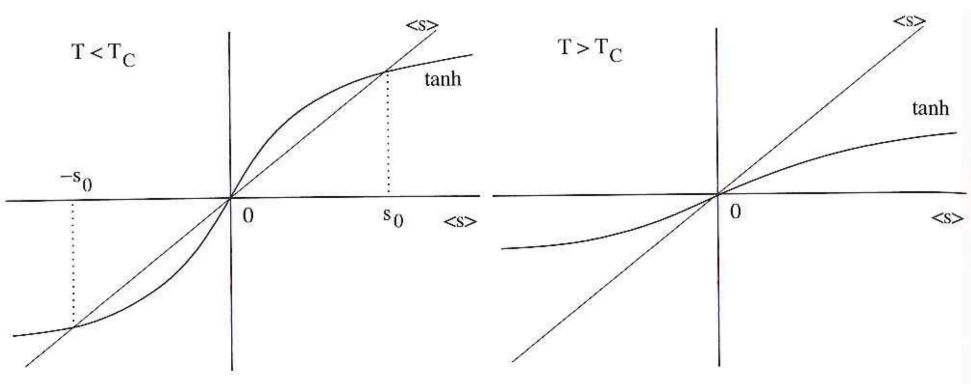
- Assume also that the external field $H = 0$, and the **effective field** becomes (z is the number of nearest neigbors)

$$H_{eff} = \frac{J}{\mu}\sum\langle s\rangle = \frac{zJ}{\mu}\langle s\rangle$$

- Combining this with the solution for thermal average leads to an **implicit relation** which cannot be solved analytically

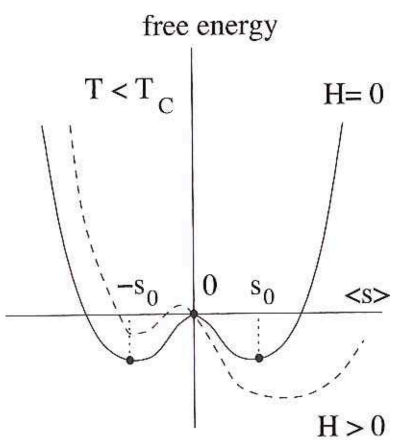
$$\langle s\rangle = \tanh(zJ\langle s\rangle/k_B T)$$

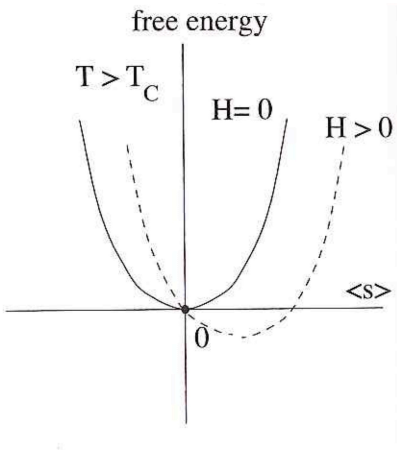
- The **numerical solution** is illustrated graphically below; the solutions correspond to intersection points



Note: The two solutions disappear as T is increased

- ***Paramagnetic phase*** for <s> = 0, magnetization (up/down) at low T



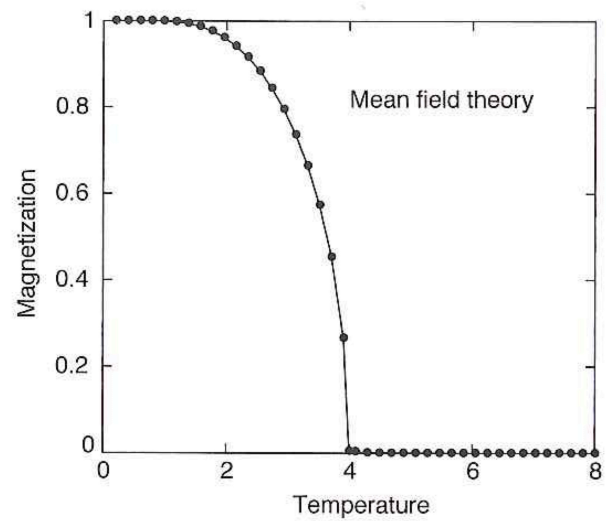


- At low temperature, the magnetized phases ($s \neq 0$) have a lower energy than the paramagnetic solution → **MFT predicts a ferromagnet!**
- If $H \neq 0$ (external field switched on), the symmetry for different magnetizations is lost and there is only one global minimum
- The solutions (as a function of T) can be computed numerically by solving the following equation with any of the **root finding methods**

$$f(\langle s \rangle) \equiv \langle s \rangle - \tanh(zJ\langle s \rangle / k_B T) = 0$$

- Function smooth and easy to differentiate analytically → ***Newton-Raphson method***

- The **positive branch** ($\langle s \rangle \geq 0$) of the mean field solution is shown on the right for a square lattice (four neighbors); J/k_B set to unity
- Spontaneous magnetization $M > 0$ at low T, ferromagnet
- At $T \geq 4$ disorder dominates and the system is paramagnetic with $M = 0$



- The slope of M changes discontinuosly as $M \rightarrow 0$ and the transition between the two phases is abrupt at the ***critical temperature, T_c***
- MFT predicts $T_c = 4$ (in units of J/k_B)
- This is an example of a ***second-order phase transition***; the spontaneous magnetization is the **order parameter** which tells the phase of the system in a given temperature
- M vanishes singularly at T_c, and the form of this singularity can be obtained analytically using a Taylor series ($\tanh(x) \approx x - x^3/3$) for small $\langle s \rangle$

$$\langle s \rangle \approx \frac{zJ\langle s \rangle}{k_B T} - \frac{1}{3}\left(\frac{zJ\langle s \rangle}{k_B T}\right)^3$$

- This has the solutions <*s*> = 0 and (**check**)

$$\langle s \rangle = \sqrt{\frac{3}{T}\left(\frac{zJ\langle s \rangle}{k_B T}\right)^3}\left(\frac{zJ}{k_B} - T\right)^{1/2} \sim (T_c - T)^{\beta}$$

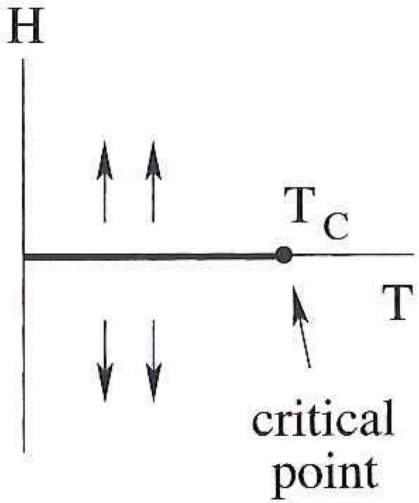
- The last step uses $T_c = zJ/k_b T = 4$ and $\beta = ½$ according to MFT
- The parameter β is a ***critical exponent*** (compare with percolation)
- A second-order phase transition is characterized by power law behavior for many properties → critical exponents
- The MFT prediction for power law behavior is correct but the value of the critical exponent is wrong

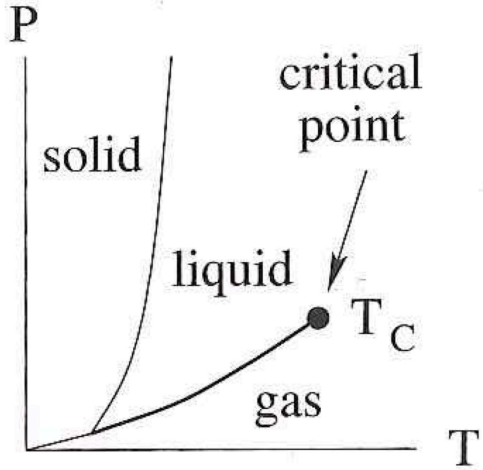
First-order phase transitions

- So far, we have considered the Ising model **without the external field** → *second-order phase transitions*
- We can observe a ***first-order transition*** if we switch the magnetic field on
- Monte Carlo scheme needs to include the energy gained/lost to the field

$$E = -J \sum_{\langle ij \rangle} s_i s_j - \mu H \sum_i s_i$$

- There are now **two independent variables**, T and H → larger phase diagram to explore
- The ***discontinuous change*** of M as the external field is varied is an indicator of a first-order transition (low T, below critical point, *next slide*)
- Simultaneous reversal of all of the spins so that $M \rightarrow -M$
- The effect of temperature is to smear the effect (above T_c)
- Sometimes the transition is associated with ***hysteresis*** → system gets temporarily trapped into a **metastable state**





Phase diagrams:

1. *H-T* plane for a ferromagnet
2. *P-T* plane of typical pure substance

- The **ferromagnetic system** has two distinct phases at low *T* corresponding to ±*M*; these are divided by a thick line that denotes a first-order transition and ends at the critical point
- Spontaneous magnetization (first-order transition) vanishes at T_c, and this has an analogy in a liquid-gas system; it is possible to go from one phase to another continuously (**note:** this can be used as a D-tour)
- The relevant variable in a ***liquid-gas system*** are **pressure** (← external field), **temperature**, and **density** (← magnetization); there are singularities at T_c which are described by ***power laws and critical exponents***
- The values of critical exponents are the same as for the Ising model (?!)

Scaling

- Let us start with the **mean field solution** for magnetization (H = 0) and add the external field

$$\langle s \rangle = \tanh[(zJ\langle s \rangle + \mu H)/k_B T]$$

- For small values of magnetization and external field near the critical point we can use the **expansion** (two terms for magnetization, one for the field)

$$\langle s \rangle \approx \frac{zJ\langle s \rangle}{k_B T} - \frac{1}{3}\left(\frac{zJ\langle s \rangle}{k_B T}\right)^3 + \frac{\mu H}{k_B T}$$

- This is an ***equation of state***, order parameter m depends on the thermodynamic variables t and h via this implicit relation (b and u are positive constants)

$$\boxed{h = btm + um^3, \qquad h \equiv \mu H/J, \qquad m \equiv \langle s \rangle, \qquad t \equiv 1 - \frac{zJ}{k_B T} = \frac{T - T_c}{T_c}}$$

- Focusing on various powers of *m*, *t*, and *h*, we can show for any λ (check)

$$m(\lambda^{1/2}t, \lambda^{3/4}h) = \lambda^{1/4}m(t,h)$$

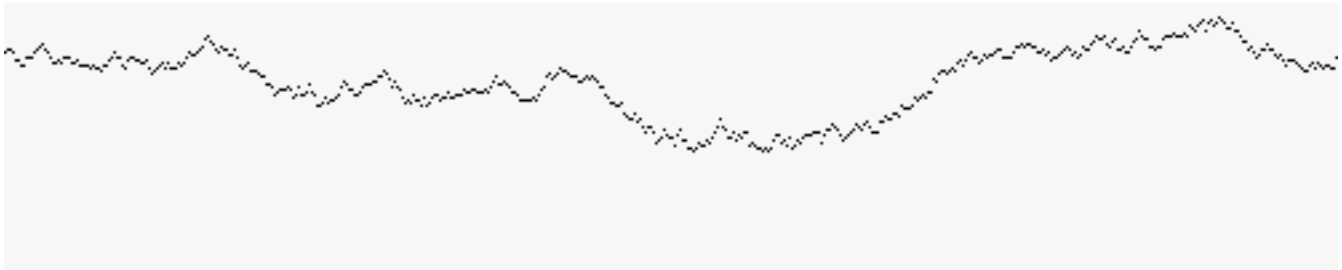
- For example, test this with $\lambda = 16$
- Functions of two variables which satisfy such an equation can be expressed as a function of a ***single*** variable
- Choose $\lambda = |t|^{-2}$ (note: cases "±" for $t < 0$ and $t > 0$)

$$m(t,h) = |t|^{1/2}m\left(\pm 1, \frac{h}{|t|^{3/2}}\right) = |t|^{1/2}f_{\pm}\left(\frac{h}{|t|^{3/2}}\right)$$

- This **scaling** results in that $m(t,h)$ is is a function of a single composite variable $h/|t|^{3/2}$; this is based on the mean field theory (MFT)
- **General scaling form**, beyond MFT

$$m(t,h) = |t|^{\beta}f_{\pm}\left(\frac{h}{|t|^{\beta\delta}}\right)$$

- For MFT, $\beta = ½$ and $\delta = 3$; the correct values are $\beta = ⅛$ and $\delta = 15$

- Scaling property of thermodynamic quantities near the critical point provides a connection between various power laws; this is understood in terms of the ***renormalization group (RG)*** theory of critical phenomena
- **RG** refers to a mathematical apparatus that allows systematic investigation of the changes of a physical system as viewed at different ***distance scales***; the system at one scale will appear to consist of self-similar copies of itself when viewed at a smaller scale
- Similarly as for magnetization, it can be shown for ***magnetic susceptibility*** (critical exponent γ = 7/4, practices)

$$\chi(t,h) = |t|^{-\gamma} g_{\pm}\left(\frac{h}{|t|^{\beta\delta}}\right), \qquad \left[\chi = \frac{(\Delta M)^2}{k_B T}\right]$$